Few-Shot Visual Recognition for Biodiversity Monitoring

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Abstract

Automated species identification is fundamental to scalable biodiversity monitoring, yet most deep learning classifiers require hundreds or thousands of labeled images per species—a prohibitive requirement for the vast majority of Earth's estimated 8.7 million species. We present **TaxoNet**, a few-shot visual recognition framework designed for fine-grained species classification in ecological monitoring contexts. TaxoNet combines a vision transformer backbone pre-trained on large-scale ecological imagery with a taxonomy-aware meta-learning strategy that exploits the hierarchical structure of biological classification (kingdom through species). A novel *taxonomic prototypical network* constructs class prototypes at multiple taxonomic ranks and fuses them through learned rank-attention weights, enabling robust identification from as few as 1–5 reference images per species. We evaluate TaxoNet on four biodiversity datasets spanning birds, insects, plants, and marine organisms, achieving state-of-the-art 1-shot accuracy of 78.4% and 5-shot accuracy of 91.2% on held-out species. We further demonstrate integration with citizen science platforms, where TaxoNet assists non-expert observers in identifying species with 94.7% accuracy when combined with geographic and phenological priors. All models and datasets are released through the Zoo Open Science Network.

Keywords: few-shot learning, fine-grained classification, biodiversity monitoring, taxonomic hierarchy, citizen science, species identification

Contents

1 Introduction

Biodiversity monitoring at global scales requires automated species identification systems capable of operating across the full breadth of taxonomic diversity [Christin et al., 2019]. Camera traps, acoustic sensors, citizen science platforms, and environmental DNA (eDNA) sampling generate petabytes of observational data annually, far exceeding manual identification capacity [Steenweg et al., 2017].

Deep learning has revolutionized image-based species identification, with convolutional neural networks achieving expert-level performance on well-studied taxa such as birds [Van Horn et al., 2015], plants [Goëau et al., 2022], and mammals [Norouzzadeh et al., 2018]. However, these successes depend on large labeled training datasets—a critical limitation given that:

(i) The majority of species have fewer than 50 available images in public databases [Lorieul et al., 2022].

(ii) Newly discovered species (approximately 18,000 per year [Costello et al., 2013]) have no training data at all.

(iii) Rare and cryptic species—often those of greatest conservation concern—are systematically underrepresented in image databases.

(iv) Intraspecific visual variation (sexual dimorphism, juvenile plumage, seasonal morphs) multiplies the effective number of required examples.

Few-shot learning addresses this data scarcity by training models to recognize novel classes from very few examples [Wang et al., 2020]. While few-shot methods have advanced rapidly in computer vision, their application to ecological species identification introduces unique challenges:

- **Fine-grained discrimination**: Species within the same genus can differ by subtle visual features (wing venation, petal shape, scale patterns) that require high-resolution feature extraction.
- **High inter-class similarity**: Closely related species form dense clusters in feature space, making prototype-based methods fragile.
- **Domain shift**: Training images from curated databases differ substantially from field photos taken by citizen scientists under variable conditions.
- **Hierarchical structure**: Biological taxonomy provides a natural hierarchy that standard few-shot methods ignore.

1.1 Contributions

We introduce **TaxoNet**, a few-shot species recognition system with the following contributions:

1. A **taxonomy-aware prototypical network** that constructs multi-rank prototypes (family, genus, species) and fuses them through learned attention, enabling few-shot identification to leverage phylogenetic structure.
2. A **hierarchical meta-learning** training strategy with taxonomic episode sampling that ensures the model learns discriminative features at every rank.
3. A **domain-adaptive pre-training** pipeline using 28 million ecological images from iNaturalist, GBIF, and museum collections.

4. Integration with **geographic and phenological priors** that improve practical field identification accuracy by constraining the candidate species set.

5. Comprehensive evaluation across **four major taxonomic groups** (birds, insects, plants, marine organisms) with separate validation on **citizen science field data**.

1.2 Zoo Labs Foundation Context

This work supports Zoo Labs Foundation's mission of open, decentralized science infrastructure. TaxoNet is designed for deployment on the Zoo citizen science platform and integrates with the Foundation's decentralized data pipelines. Model weights and training code are released under the Apache 2.0 license, and the evaluation benchmark is maintained as a community resource through the Zoo Open Science Network at `zips.zoo.ngo`.

2 Related Work

2.1 Species Identification with Deep Learning

The past decade has seen dramatic advances in automated species identification. iNaturalist's computer vision system classifies over 76,000 species [Van Horn et al., 2021]. eBird's Merlin app achieves high accuracy for North American birds using audio and image inputs [Kahl et al., 2021]. PlantNet identifies over 38,000 plant species from photographs [Goëau et al., 2022]. These systems rely on extensive labeled datasets, with performance degrading sharply for species with fewer than 100 training images.

2.2 Few-Shot Learning

Few-shot learning aims to classify novel classes from K labeled examples per class (a "K-shot" setting). Metric-based approaches learn an embedding space where classification reduces to nearest-neighbor comparison: matching networks [Vinyals et al., 2016], prototypical networks [Snell et al., 2017], and relation networks [Sung et al., 2018]. Optimization-based methods learn initialization points for rapid adaptation: MAML [Finn et al., 2017] and its variants. More recently, large pre-trained models with in-context learning have shown strong few-shot capabilities [Brown et al., 2020].

2.3 Hierarchical Classification

Taxonomic hierarchy has been exploited in traditional species classification through hierarchical softmax [Deng et al., 2014], label embedding with taxonomy trees [Akata et al., 2015], and curriculum learning from coarse to fine [Bengio et al., 2009]. Chang et al. [2021] incorporate hierarchical labels into contrastive learning. However, hierarchical few-shot learning remains underexplored, with only Li et al. [2019] and Liu et al. [2020] addressing it in non-ecological contexts.

2.4 Fine-Grained Recognition

Fine-grained visual classification distinguishes subordinate categories (species, cultivars, model variants). Key approaches include bilinear pooling [Lin et al., 2015], part-based attention [Zheng et al., 2017], and feature recalibration [Hu et al., 2018]. Vision transformers (ViTs) have become dominant due to their ability to capture global context [Dosovitskiy et al., 2021, He et al., 2022]. Few-shot fine-grained recognition remains challenging due to the high intra-class and low inter-class variance [Wei et al., 2019].

3 Problem Formulation

3.1 Task Definition

We consider the standard N-way K-shot classification problem. Given a *support set* $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^{N\times K}$ containing K labeled images for each of N novel species, and a *query set* $\mathcal{Q} = \{x_j\}_{j=1}^{Q}$ of unlabeled images, the goal is to classify each query into one of the N species.

3.2 Taxonomic Structure

Each species s belongs to a taxonomic hierarchy:

$$\text{Kingdom} \supset \text{Phylum} \supset \text{Class} \supset \text{Order} \supset \text{Family} \supset \text{Genus} \supset \text{Species} \tag{1}$$

We denote the taxonomic label of species s at rank r as $t_r(s)$. Two species in the same genus share t_r for $r \in \{\text{kingdom}, \dots, \text{genus}\}$ but differ at species rank.

3.3 Incorporating Priors

For practical deployment, geographic coordinates (lat, lon) and observation date d provide strong priors on which species are plausible at a given location and time. We define the candidate set:

$$\mathcal{C}(\text{lat}, \text{lon}, d) = \{s : s \text{ has recorded occurrence within } r \text{ km and } \pm \Delta d \text{ days}\} \tag{2}$$

which restricts the N-way classification to geographically and temporally plausible species.

4 TaxoNet Architecture

4.1 Overview

TaxoNet consists of four components: (1) a vision transformer feature extractor, (2) a taxonomic prototype constructor, (3) a rank-attention fusion module, and (4) optional geographic/phenological prior integration.

4.2 Feature Extractor

We use a Vision Transformer (ViT-L/14) pre-trained through our domain-adaptive pipeline (Section 5). Input images are resized to 518×518 pixels and split into 14×14 patches. The transformer produces a sequence of patch embeddings, from which we extract both the `[CLS]` token embedding $\mathbf{h} \in \mathbb{R}^{1024}$ and the full patch embedding matrix $\mathbf{H} \in \mathbb{R}^{1369\times 1024}$.

For fine-grained discrimination, we apply a learned spatial attention module that identifies discriminative image regions:

$$\mathbf{a} = \text{softmax}(\mathbf{W}_a \mathbf{H}^\top) \in \mathbb{R}^{M\times 1369} \tag{3}$$

where $M = 4$ attention heads focus on different discriminative parts. The attended feature is:

$$\mathbf{f} = [\mathbf{h} \| \text{vec}(\mathbf{aH})] \in \mathbb{R}^{1024+4\times 1024} = \mathbb{R}^{5120} \tag{4}$$

projected to a 512-dimensional embedding via a linear layer.

4.3 Taxonomic Prototypical Network

Given the support set $\mathcal{S}$ for an N-way K-shot episode, we construct prototypes at three taxonomic ranks: family (r_F), genus (r_G), and species (r_S).

Species prototype. The standard prototypical network prototype:

$$\mathbf{c}_{r_S}^{(s)} = \frac{1}{K} \sum_{(x_i, y_i) \in \mathcal{S}: y_i = s} \phi(x_i) \tag{5}$$

where $\phi(\cdot)$ is the feature extractor.

Genus prototype. Aggregating all species in the same genus:

$$\mathbf{c}_{r_G}^{(g)} = \frac{1}{|\{s : t_{r_G}(s) = g\}|} \sum_{s: t_{r_G}(s) = g} \mathbf{c}_{r_S}^{(s)} \tag{6}$$

Family prototype. Similarly aggregating across genera in the same family:

$$\mathbf{c}_{r_F}^{(f)} = \frac{1}{|\{g : t_{r_F}(g) = f\}|} \sum_{g: t_{r_F}(g) = f} \mathbf{c}_{r_G}^{(g)} \tag{7}$$

4.4 Rank-Attention Fusion

For a query embedding $\mathbf{q} = \phi(x)$, we compute distances to each prototype at each rank. The rank-attention module learns to weight these distances adaptively:

$$d_r(x, s) = \|\mathbf{q} - \mathbf{c}_r^{(t_r(s))}\|^2 \tag{8}$$

The attention weights depend on the query itself:

$$\boldsymbol{\alpha}(\mathbf{q}) = \mathrm{softmax}(\mathbf{W}_r \mathbf{q}) \in \mathbb{R}^3 \tag{9}$$

The fused distance is:

$$d_{\text{fused}}(x, s) = \sum_{r \in \{r_F, r_G, r_S\}} \alpha_r(\mathbf{q}) \cdot d_r(x, s) \tag{10}$$

The predicted class probability is:

$$p(y = s \mid x) = \frac{\exp(-d_{\text{fused}}(x, s))}{\sum_{s'} \exp(-d_{\text{fused}}(x, s'))} \tag{11}$$

4.5 Geographic and Phenological Priors

When location and date metadata are available, we compute a prior probability $p_{\text{geo}}(s \mid \text{lat}, \text{lon}, d)$ from species range maps and phenological data. This prior is combined with the visual prediction via:

$$p_{\text{final}}(s \mid x, \text{lat}, \text{lon}, d) \propto p(y = s \mid x)^{1-\beta} \cdot p_{\text{geo}}(s \mid \text{lat}, \text{lon}, d)^{\beta} \tag{12}$$

where $\beta = 0.3$ balances visual and geographic evidence. This effectively re-ranks predictions by geographic plausibility.

5 Domain-Adaptive Pre-Training

Standard ImageNet pre-training provides suboptimal features for fine-grained ecological recognition due to domain mismatch (controlled studio photos vs. field conditions) and category mismatch (1,000 general categories vs. thousands of visually similar species).

5.1 Pre-Training Data

We assemble **EcoVision-28M**, a dataset of 28 million ecological images from:

- **iNaturalist 2021**: 10 million images across 10,000 species [Van Horn et al., 2021].
- **GBIF media**: 8 million specimen and field images.
- **Museum collections**: 5 million digitized specimen photographs from 14 natural history museums.
- **Camera trap archives**: 3 million images from Wildlife Insights and LILA BC [Beery et al., 2021].
- **Citizen science platforms**: 2 million images from eBird, BugGuide, and Reef Life Survey.

5.2 Self-Supervised Pre-Training

We pre-train ViT-L/14 using DINOv2 [Oquab et al., 2024] on EcoVision-28M for 300,000 iterations with batch size 1,024. The self-supervised objective learns representations that capture fine-grained visual structure without requiring species labels.

5.3 Supervised Fine-Tuning

After self-supervised pre-training, we fine-tune on the labeled subset (iNaturalist 2021, 10M images, 10,000 species) using a hierarchical classification head with losses at family, genus, and species level:

$$\mathcal{L}_{\text{pretrain}} = \mathcal{L}_{\text{species}} + 0.3\mathcal{L}_{\text{genus}} + 0.1\mathcal{L}_{\text{family}} \tag{13}$$

6 Hierarchical Meta-Learning

6.1 Episode Construction

Standard meta-learning samples episodes uniformly. We introduce **taxonomic episode sampling** that ensures episodes contain discriminative challenges at each rank.

For each training episode, we:

1. Sample a focal family f with probability proportional to its species richness.
2. From f, sample 2–4 genera.
3. From each genus, sample 2–5 species.
4. Include 1–2 "distractor" species from the same order but different family.

This ensures the model must discriminate at multiple taxonomic levels within each episode, unlike uniform sampling which often produces episodes of distantly related, trivially distinguishable species.

6.2 Multi-Rank Loss

The training loss combines classification losses at each rank:

$$\mathcal{L}_{\text{meta}} = \mathcal{L}_{\text{CE}}^{(r_S)} + \gamma_G \mathcal{L}_{\text{CE}}^{(r_G)} + \gamma_F \mathcal{L}_{\text{CE}}^{(r_F)} + \lambda \mathcal{L}_{\text{triplet}} \tag{14}$$

where $\mathcal{L}_{\text{CE}}^{(r)}$ is the cross-entropy loss for classification at rank r, $\mathcal{L}_{\text{triplet}}$ is a taxonomic triplet loss that pushes apart embeddings of different genera while pulling together congeneric species, and $\gamma_G = 0.3$, $\gamma_F = 0.1$, $\lambda = 0.2$ are weighting coefficients.

6.3 Taxonomic Triplet Loss

We define:

$$\mathcal{L}_{\text{triplet}} = \sum_{(a,p,n)} \max(0, \|\phi(x_a) - \phi(x_p)\| - \|\phi(x_a) - \phi(x_n)\| + m(a, n)) \tag{15}$$

where (a, p, n) is an anchor-positive-negative triplet, and the margin $m(a, n)$ scales with taxonomic distance:

$$m(a, n) = m_0 \cdot (1 - \text{sim}_{\text{tax}}(a, n)) \tag{16}$$

Here $\text{sim}_{\text{tax}}(a, n) \in [0, 1]$ is the fraction of shared taxonomic ranks, and $m_0 = 0.5$ is the base margin.

6.4 Training Protocol

- **Meta-training set**: 8,000 species (disjoint from evaluation species).
- **Meta-validation set**: 1,000 species.
- **Episodes per epoch**: 10,000.
- **Optimizer**: AdamW, learning rate 5×10^{-5}, cosine schedule over 100 epochs.
- **Episode configuration**: 10-way K-shot, $K \in \{1, 5\}$, with 15 queries per class.

7 Evaluation

7.1 Datasets

We evaluate on four domain-specific benchmarks:

Table 1: Evaluation datasets. "Novel" species have no overlap with training data.

Dataset	Taxon	Novel species	Images	Source
ZooBirds	Aves	412	24,720	eBird, Macaulay
ZooInsects	Insecta	638	31,900	iNaturalist, BugGuide
ZooPlants	Plantae	524	26,200	PlantNet, herbaria
ZooMarine	Mixed marine	293	14,650	Reef Life Survey

Citizen science field test. We additionally evaluate on 2,847 field photographs submitted by 312 citizen science volunteers through the Zoo platform, covering 187 species across all four taxa. Images are accompanied by GPS coordinates and dates.

7.2 Baselines

- **ProtoNet** [Snell et al., 2017]: Standard prototypical network with ResNet-50 backbone.
- **MAML** [Finn et al., 2017]: Model-agnostic meta-learning with 5 inner-loop steps.
- **DeepEMD** [Zhang et al., 2020]: Earth mover's distance matching for few-shot.
- **FEAT** [Ye et al., 2020]: Set-to-set function with transformer adaptation.
- **P>M>F** [Hu et al., 2022]: Pre-train, meta-learn, fine-tune pipeline.

- **DINOv2+kNN**: DINOv2 features with k-nearest-neighbor classification (no meta-learning).

All baselines use ViT-L/14 backbone for fair comparison except where architecturally infeasible.

7.3 Results: Standard Few-Shot

Table 2: Few-shot classification accuracy (%) on novel species. Mean ± 95% CI over 10,000 episodes.

	ZooBirds		**ZooInsects**		**ZooPlants**		**ZooMarine**	
Method	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
ProtoNet	62.3	79.8	54.7	73.2	58.1	76.4	60.9	78.1
MAML	64.1	81.3	56.2	74.8	59.7	77.9	62.4	79.6
DeepEMD	67.8	84.1	59.3	78.6	63.2	81.2	65.7	82.3
FEAT	69.2	85.4	61.1	80.3	65.1	82.7	67.3	83.8
P>M>F	72.6	87.9	64.8	83.1	68.4	85.3	70.1	86.2
DINOv2+kNN	71.3	86.2	63.1	81.4	66.7	83.9	68.8	84.7
TaxoNet	**80.1**	**92.4**	**74.2**	**89.1**	**78.9**	**91.6**	**80.4**	**91.7**

TaxoNet achieves the highest accuracy across all datasets and shot settings. The largest improvement over the next-best method (P>M>F) is on ZooInsects (+9.4% 1-shot), consistent with insects having the deepest taxonomic hierarchies and most confusable species pairs.

7.4 Results: Citizen Science Field Test

Table 3: Field identification accuracy (%) on citizen science photographs. "+Geo" adds geographic and temporal priors.

Method	**Top-1**	**Top-3**	**Top-5**	**Top-1 +Geo**
iNaturalist CV	82.3	91.4	94.1	87.6
DINOv2+kNN	76.8	88.2	92.3	83.1
P>M>F (5-shot)	79.4	89.7	93.2	85.8
TaxoNet (5-shot)	**84.7**	**93.1**	**96.2**	**94.7**

With geographic priors, TaxoNet achieves 94.7% top-1 accuracy on citizen science data, exceeding iNaturalist's trained classifier on overlapping species. This demonstrates the practical value of combining few-shot visual recognition with ecological metadata.

7.5 Ablation Studies

Table 4: Ablation study on ZooBirds (5-way 5-shot accuracy %).

Configuration	**Accuracy**
Full TaxoNet	**92.4**
– Taxonomic prototypes (species only)	88.7
– Rank attention (uniform weights)	90.1
– Domain-adaptive pre-training (ImageNet init)	86.3
– Taxonomic episode sampling (uniform)	89.8
– Spatial attention (CLS token only)	90.6
– Taxonomic triplet loss	91.2

Domain-adaptive pre-training provides the largest single contribution (+6.1%), followed by taxonomic prototypes (+3.7%) and taxonomic episode sampling (+2.6%).

7.6 Rank-Attention Analysis

Table 5: Learned rank-attention weights α_r for different query difficulty levels.

Query difficulty	α_{family}	α_{genus}	α_{species}
Easy (inter-family)	0.62	0.24	0.14
Medium (inter-genus)	0.18	0.49	0.33
Hard (intra-genus)	0.07	0.28	0.65

The rank-attention module adaptively shifts weight toward species-level prototypes for hard (intra-genus) queries and toward family-level prototypes for easy (inter-family) queries.

8 System Design for Citizen Science

8.1 Deployment Architecture

TaxoNet is deployed as a REST API service behind a geographic routing layer. When a citizen scientist submits a photograph:

1. **Candidate filtering**: Geographic and temporal priors reduce the candidate species set to 50–200 plausible species.

2. **Feature extraction**: The ViT backbone extracts visual features in approximately 45 ms on an NVIDIA T4 GPU.

3. **Few-shot matching**: Taxonomic prototypical matching against the candidate set runs in approximately 12 ms.

4. **Result presentation**: Top-5 predictions with confidence scores and diagnostic field marks are returned to the user.

Total inference latency is under 100 ms per image.

8.2 Prototype Management

Species prototypes are computed offline from curated reference images and stored in a vector database indexed by taxonomy and geography. New species can be added by computing and inserting their prototype embeddings—no model retraining is required.

8.3 Active Learning Loop

Citizen science identifications that disagree with model predictions are flagged for expert review. Confirmed identifications update prototypes:

$$\mathbf{c}^{(s)}_{\text{new}} = (1-\eta)\mathbf{c}^{(s)}_{\text{old}} + \eta \cdot \phi(x_{\text{confirmed}}) \tag{17}$$

where $\eta = 0.01$ is the update rate.

9 Discussion

9.1 Why Taxonomy Matters

The success of taxonomic prototypes reflects a biological reality: species within the same genus share morphological features due to common ancestry. By explicitly modeling this hierarchical structure, TaxoNet leverages phylogenetically informative features even when species-level examples are scarce. The rank-attention mechanism acts as a soft decision tree, routing queries to the appropriate level of taxonomic resolution based on visual evidence quality.

9.2 Limitations

Cryptic species. Species complexes that are morphologically indistinguishable cannot be resolved by visual features alone. Integration with molecular data (eDNA) or acoustic features is needed.

Phenological variation. Some species change appearance dramatically across seasons. Explicit temporal prototype modulation would be more principled than our current update mechanism.

Geographic bias. Pre-training data is biased toward North America and Europe. Performance on tropical and Southern Hemisphere species may be lower due to representation gaps.

9.3 Ethical Considerations

- **Poaching risk**: We implement geographic fuzzing for threatened species locations.
- **Over-reliance**: We display confidence calibration and encourage expert confirmation for low-confidence predictions.
- **Data sovereignty**: We follow the Nagoya Protocol and CARE principles for indigenous data governance.

10 Implementation Details

10.1 Software and Hardware

TaxoNet is implemented in PyTorch 2.2 with the `timm` library for ViT models. Training was conducted on 8 NVIDIA A100 GPUs for approximately 72 hours. Inference is deployed on NVIDIA T4 GPUs behind a Kubernetes load balancer.

10.2 Data Augmentation

Training episodes apply random resized crop (scale 0.4–1.0), horizontal flip, color jitter, random Gaussian blur, random grayscale (probability 0.1), and cutout with one 64×64 hole.

10.3 Confidence Calibration

We calibrate prediction confidence using temperature scaling [Guo et al., 2017]. Post-calibration, expected calibration error (ECE) drops from 8.3% to 2.1%.

11 Future Work

1. **Multimodal fusion**: Combining visual features with audio and eDNA sequence embeddings for visually cryptic species.
2. **Continual meta-learning**: Adapting as new species are described without catastrophic forgetting.
3. **Federated training**: Training on distributed citizen science data without centralizing raw images, governed by Zoo DAO.
4. **Zero-shot identification**: Leveraging taxonomic descriptions and field guide text via vision-language models.
5. **Decentralized inference**: Deploying TaxoNet on edge devices and the Zoo decentralized compute network.

12 Conclusion

We have presented TaxoNet, a taxonomy-aware few-shot visual recognition system for biodiversity monitoring. By incorporating biological classification structure into meta-learning through taxonomic prototypical networks and rank-attention fusion, TaxoNet achieves state-of-the-art few-shot accuracy across four major taxonomic groups. Integration with geographic and phenological priors further boosts practical field identification to 94.7% accuracy. Through Zoo Labs Foundation's commitment to open science, TaxoNet's models, benchmarks, and deployment infrastructure are freely available to the conservation community.

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A Supplementary: Per-Taxon Analysis

Table 6: Detailed per-taxon performance breakdown (5-way 5-shot %).

Dataset	**Difficulty**	**TaxoNet**	**P>M>F**
ZooBirds	Inter-family	97.8	96.1
	Inter-genus	93.6	88.4
	Intra-genus	86.2	78.9
ZooInsects	Inter-family	96.4	94.2
	Inter-genus	90.1	83.7
	Intra-genus	81.3	71.6
ZooPlants	Inter-family	97.1	95.3
	Inter-genus	92.8	86.1
	Intra-genus	84.7	74.8
ZooMarine	Inter-family	97.3	95.7
	Inter-genus	92.4	87.0
	Intra-genus	85.1	76.2

TaxoNet's advantage is most pronounced for intra-genus discrimination (+7.3% to +9.9% over P>M>F), precisely the setting where taxonomic prototypes provide the most value.

B Supplementary: Prototype Visualization

Analysis of learned prototypes using t-SNE reveals clear taxonomic clustering: species prototypes cluster tightly within genera, genera form distinct sub-clusters within families, and families separate cleanly in the embedding space. The taxonomic prototypical network naturally organizes the embedding space to reflect biological relationships, emerging from the hierarchical episode sampling and multi-rank loss.

Spatial attention maps show that TaxoNet focuses on taxonomically informative features: wing bar patterns for warbler identification, venation patterns for butterfly classification, leaf margin shapes for plant identification, and operculum structure for marine gastropod classification. These attention patterns are consistent with the diagnostic characters used by human taxonomists.